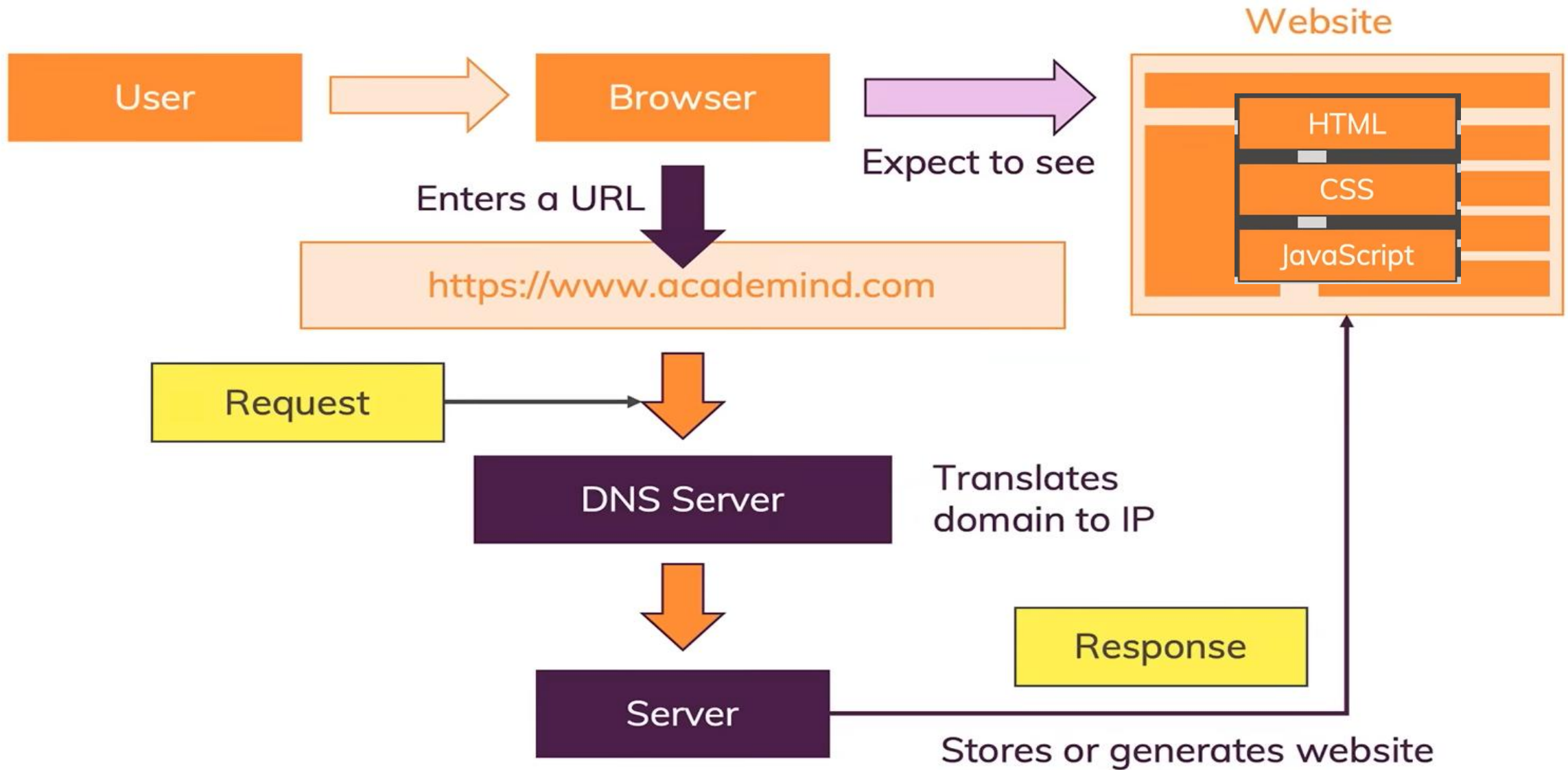


24CSA212 – Full Stack Frameworks

INTRODUCTION TO WEB DEVELOPMENT -INTRODUCTION TO FULL
STACK -INTRODUCTION TO MERN STACK -SETTING UP DEVELOPMENT
ENVIRONMENT - BASICS OF HTML, CSS, AND JAVASCRIPT -
INTRODUCTION TO MONGODB

How The Web Works



HTML

HYPERTEXT MARKUP LANGUAGE - THE WEBSITE'S STRUCTURE

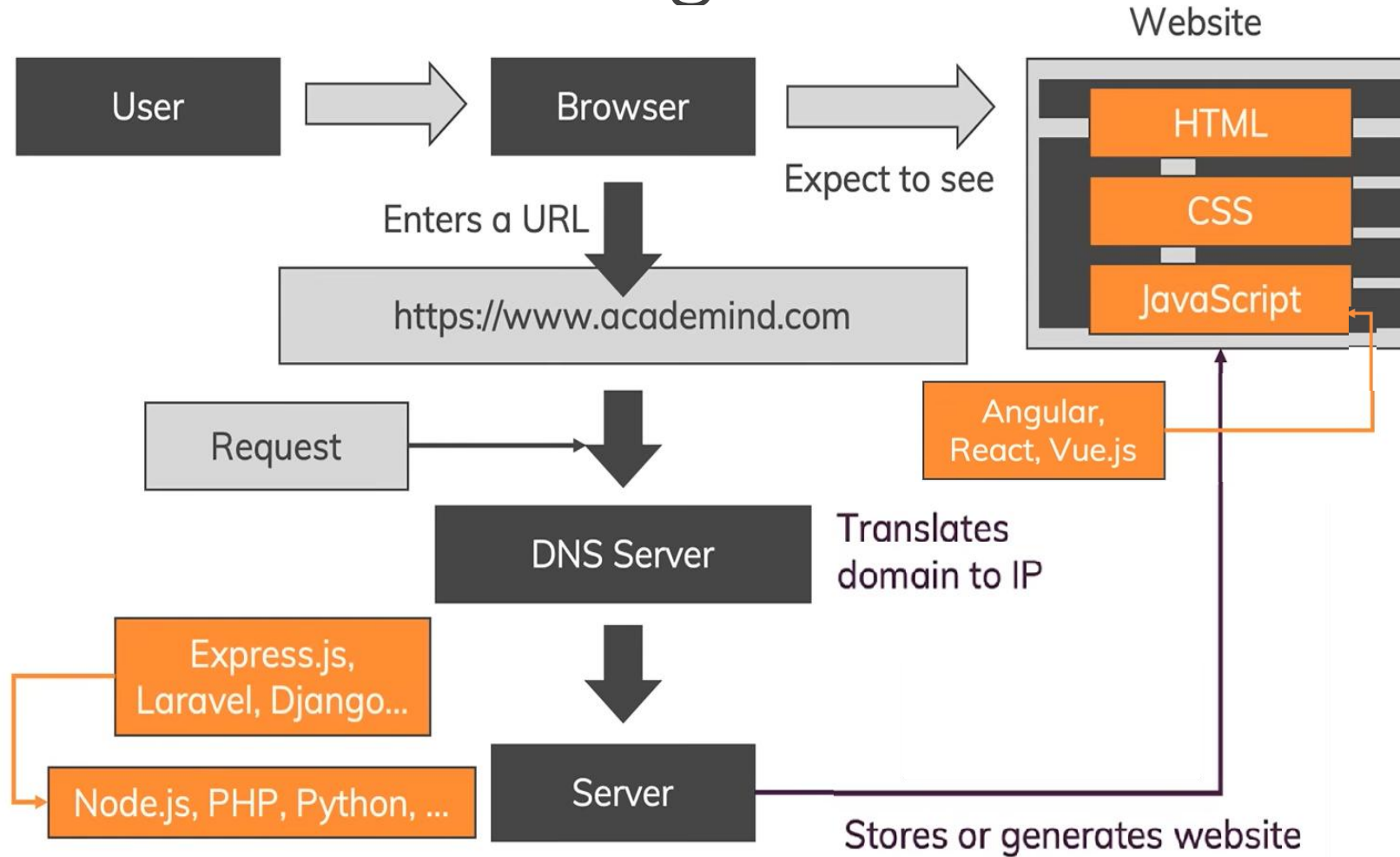
CSS

CASCADING STYLE SHEETS - MAKING IT LOOK GOOD

JAVASCRIPT

ADDING LOGIC TO A WEBSITE

More Technologies involved



Server Side Coding – Node.JS, PHP, Python ...

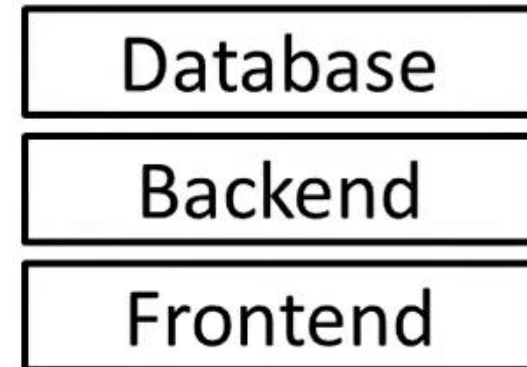
Frameworks – Packages of utility functions that also give info about how to use them
Browser side – Need not load the page each time. Tell What to change, when

Technology Stack

Technology stack is a collection of three tiers of technologies.

Full stack developer is a person who can work on frontend, backend and Database

- **Frontend technology**
- **Backend technology**
- **Database**



MERN Stack



Mongodb



Express 

Express JS is a framework which runs inside NodeJS. It handles all the requests and responses coming from the frontend to the backend.



Read

Client Side – Front end library, On the web end where the user is interacting with your website or web application, is called client side



Node.js

Node JS - Server side – where your logics will run or request or response will be done
It is a runtime environment which is used to build the application on server-end

MERN Stack

Front-end (browser)

Back-end (server)

React App



Express / Node API

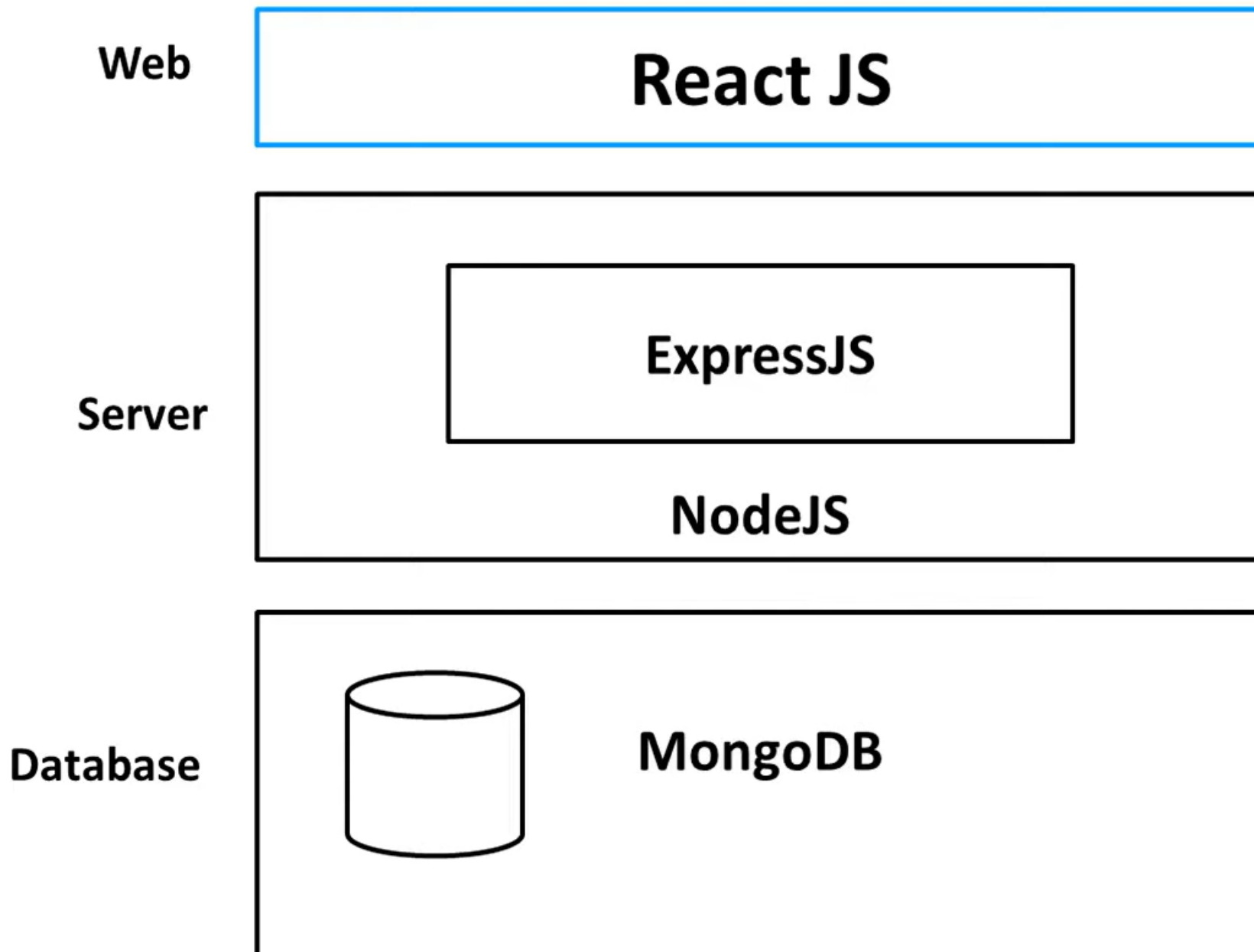


Mongodb



JSON





MongoDB and ReactJS cannot communicate with each other
Data should be shown at front end
Server comes in between in which Node JS and Express JS will run

Why use a web framework?

- 👉 Saves time
- 👉 Makes code cleaner and more organized
- 👉 Handles common tasks (like routing, error handling, etc.)

Why use **MERN Stack**?

- **MongoDB:** Stores JSON data natively. Query is also built on JavaScript.
MongoDB is a non-relational database with stores data in JSON format. Its queries are based on Javascript
- **NodeJS:** JavaScript's runtime environment to build server side application.
- **ExpressJS:** NodeJS framework that runs on the server side to handle server side queries and requests.
- **ReactJS:** JavaScript framework that runs of the client side to render data and handle events.

A single language Javascript is used to develop the entire stack. When the individual tiers communicate, there is no need of language switch, hence increased speed

When React JS requests for APIs from NodeJS or ExpressJS, then also there is no change in technology. No communication barrier.



Web Development Stack

